Jason Hong

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Computer Science and Engineering — *Sept 2024 – May 2028*

- GPA: 5.0/5.0
- Relevant Coursework: Operating Systems Engineering, Digital Systems Laboratory (FPGA), Computer Architecture, Data Structures and Algorithms, Fundamentals of Programming
- Leadership: MIT Student Advisory Group in Engineering, Associate Advisor, MIT Chinese Students and Scholars Association

Experience

Durability — Cambridge, MA
Software Engineering Intern — *June 2026 – Aug 2026*

- Built a full-stack React/Supabase coaching platform and ETL pipeline to import and normalize 48 legacy TrueCoach programs (7,600+ exercise prescriptions) into editable multiweek calendars supporting versioned publishing and athlete assignment.
- Built an end-to-end bulk roster-upload pipeline across React, Supabase/Deno, and SwiftUI, automating validation, batched invitations, email OTP, and first-login password setup.

Kyte Dynamics — Los Angeles, CA
Software Engineering Intern — *May 2025 – Sept 2025*

- Built custom bidirectional MAVLink/uORB messaging between two PX4 flight controllers over UART and validated it on hardware, implementing outbound streaming, inbound decoding, and test publication.

MIT 6.191 Computation Structures — Cambridge, MA
Lab Assistant — *Sept 2026 - present*

- Responsible for hosting office hours, tutoring students, and providing lab checkoffs.

Research

MIT Laboratory for Information & Decision Systems — Cambridge, MA
Undergraduate Researcher (UROP) | Advised by Prof. Tamara Broderick — *April 2026 – Present*

- Migrated Nomon's language-model interface to the TextSlinger API to support language-model benchmarking.
- Corrected inconsistent conditioning of letter and word probabilities in log space, reducing the simulated correction rate from 3.99% to 0.68%.
- Developed output-validity checks that revealed 33.5% of word suggestions were invalid, a failure undetected by the oracle simulator; restricting suggestions to a valid vocabulary eliminated invalid outputs and reduced simulated click load from 1.390 to 1.147.

Honors & Awards

MIT Pozen Fellowship: Awarded to select students for excellence in engineering and public service (2026)
Dr. Sherrie Emoto Mathematics Scholarship: Highest departmental honor for mathematical achievement at Fullerton College (2025)

Projects

RISC-V Neural Network Inference Accelerator | *Minispec/BSV, C, RISC-V* — Spring 2026

- Optimized a pipelined processor and MNIST inference stack, achieving an **80**× improvement over baseline.
- Extended the ISA with custom instructions and pipelined SIMD multiplication; cut clock period from **1072 ps to 529 ps** via a Kogge-Stone adder and by re-encoding ALU branch output as a single comparison bit, and added a specialized iCache to reduce CPI.

Technical Skills

Languages: C/C++, Python, React, Typescript Swift, SystemVerilog, Bash
Tools & Frameworks: Linux/Unix, Git, Neovim, SwiftUI, React, Supabase, CMake, MAVLink
Interests: Computer Architecture, Verified Software Engineering, OS Kernels, Performance Engineering